

PREVALENCE OF HEPATITIS B AND C IN BLOOD DONOR AT FAISALABAD(1) (1).docx

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²Prevalence of Hepatitis B and C in Blood Donor, at Faisalabad.

INRODUCTION

The term "hepatitis," rooted in Latin, refers to inflammation of the liver. Today, viral hepatitis remains a significant global health threat, especially in developing Asian countries, where it leads to severe liver diseases and sometimes liver cancer. Most cases arise from one primary virus, but several types can be involved; it can appear as acute or chronic, potentially causing cirrhosis or cancer. While autoimmune conditions and drugs contribute, viruses are the main cause, and types B and C are particularly deadly, often resulting in cancer or end-stage liver failure.(Nazir, Awais et al. 2025).

Viral hepatitis, especially hepatitis B remains a significant global health issue, impacting million around the world. The WHO's 2024 Global Hepatitis Report indicates that ⁵more than 254 million people live with chronic HBV ,and about 1.2 million new cases arise annually of 2022 .(Altaf, Ali et al.)

The worldwide burden of HBV infection varies significantly by region. ¹¹The Western Pacific and African regions carry the heaviest load, with 116 million and 81 million people living with chronic infections, ²respectively. The Eastern Mediterranean region (which includes Pakistan) reports around 60 million cases, followed by Southeast Asia 18 million, Europe 14 million, and the Americas 5 million.²(Waheed, Shahid et al. 2025)

Pakistan and Egypt represent 80% of viral hepatitis cases in the Eastern Mediterranean Region (EMR). In Pakistan, HBV infect 2.5% of the population. (Bhatti, Junaaid et al. 2022)

The World Health estimate global hepatitis C virus prevalence is about 1%, though this average hides stark regional differences. ⁹North Africa, the Middle East, and South Asia report some of the highest rates, particularly in Egypt and Pakistan. Within South Asia, Pakistan stands out with exceptionally high HCV prevalence—around 5-6% among adults, affecting roughly 9-10 million people.(Waheed, Khokhar et al. 2024)

Pakistan bears one of the world's heaviest HCV burdens, with a 4.3% estimated viral infections prevalence in 2021 – affecting 8.8 million people and accounting for almost 18% of the global HCV cases.(Niyomsri, Lim ²⁸ et al. 2026)

Hepatitis B and hepatitis C infections demand serious international attention due to their startling high infection rates. Chronic liver infection impact more than 200 million people globally. Around ⁵ 600,000 people worldwide die from the acute or chronic effect of year.(Arooj, Hayat et al. 2025).

Pakistan display particularly alarming epidemiological picture of ²¹ viral hepatitis. Hepatitis B and hepatitis C virus infections affect around 15 million people in Pakistan. Every day in Pakistan, an estimated 325 individuals pass away from viral hepatitis.(Mustafa, Kow et al. 2021)

Acknowledging the severity ² of this threat the WHO Global Health Sector Strategy (2016-2030) has urged the elimination of viral hepatitis as a public health issue by 2030.(Waheed, Shahid ²⁶ et al. 2025). The World Health Organization (WHO) has established bold 2030 targets to cut HBV incidence by 90% and mortality by 65%, driven by broader vaccination and treatment availability. While vaccination remains the strongest tool for managing HBV, only a fraction of infected individuals worldwide recognize their status, and even fewer get treatment.(Aziz, Akbar et al. 2024)

Transfusions ¹⁶ of blood or plasma-derived products were associated with a considerable risk of acquiring hepatitis B and hepatitis C in Pakistan, prior to the establishment of donor screening for hepatitis.(Arshad, Arif et al. 2021)

Globally, over ⁷ 17.4 million units of blood are donated annually. ⁷ In Pakistan more than 1.5 million pints are collected each year with replacement donor making up 65%, voluntary donor 25% and professional donor 10%.(Ghazanfar, Hassan et al. 2022)

The WHO places strong emphasis on routine screening ³ of all blood donations—a standard now required and widely practiced in most blood bank. Still transfusion-transmitted infections (TTIs) remain a major risk, especially in low- and middle-income countries with constrained healthcare resources and testing infrastructure. Worldwide TTI rates in donated blood differ widely: HBV from 0.03% to 3.70%, HCV from 0.02% to 1.03%. Of these, hepatitis B virus (HBV) is the leading TTI among donors, showing a consistent rise in prevalence in recent years.(Iqbal, Farhad et al. 2026)

¹⁸ Hepatitis B virus spread mainly through contact with infected blood and body fluids. To address this danger, the WHO for years that ³ all donated blood undergo screening for key transfusion-transmissible infections (TTI's) such as hepatitis viruses. (Waheed, Shahid et al. 2025)

The prevalence of HBV is influenced by factors like socioeconomic status, geographic location, and levels of public awareness. It spreads primarily through blood transfusions, sharing infections sterilized razors for shaving, poor dental hygiene practices, reusing syringes, and unprotected sexual contact with infected people. (Almas, Tariq et al. 2024)

Hepatitis C virus (HCV), related to genus of Hepacivirus, is a frequent cause of chronic liver diseases. Transmissions primarily take place through the contaminated blood products, unprotected sex, infected barber tools, tattooing, and exposure to infected body fluids. HCV serves as the leading trigger for hepatocellular carcinoma (HCC), a common tumor and one of the top causes of cancer-related deaths worldwide. When accompanied by inflammation, hepatocyte damage sparks regeneration and fibrosis, which can ultimately lead to HCC and liver cirrhosis. (Saleem, Aslam et al. 2022).

According to published studies ¹⁵ in Pakistan, the prevalence of transfusion-transmissible infections (TTIs) among blood donor stood at 5.8% in 2016, dropping to 3.72% in 2019 before rising slightly to 4.61% in 2021. The leading ²⁴ TTI was HCV, followed by HBV, syphilis and HIV. (Sabir, Ghafoor et al. 2023)

¹ The Ministry of National Health Services, Regulations and Coordination (MoNHSR&C), Government of Pakistan, has released Safe Blood Transfusion Guidelines (SBTG) mandating tests for five key TTI pathogens prior to any transfusion.

Blood donation can as yet transmit these infections, above all in Faisalabad, where screening practices differ widely between urban and rural blood banks. Risks are heightened by reusing unsterilized equipment, transfusions performed by unqualified staff, poor comprehension of infection control, and dependence on family replacement or paid donors. ¹ Strict screening and adherence to safe blood transfusion protocols are essential to single out and exclude individuals with subclinical infections. Initiatives like regional blood centers (RBCs), heightened public awareness, and required testing for central transfusions infections have advanced blood safety, the prevalence of these infections among donors continues to raise alarms. Some degree of caution is warranted in asserting that Faisalabad, the provincial capital of Punjab, grapples with extra hurdles from its underdeveloped healthcare system,

decreased literacy proportions, and widespread poverty. This gives rise to the conclusion that, these socio-demographic considerations have a bearing on late diagnosis, limited awareness of viral infections, and greater vulnerability to transmission.¹² This study aims to weigh the prevalence of HBV, and HCV, infections among healthy blood donors in Faisalabad, Pakistan. By determining baseline infection rates and pinpointing related risk factors, the results aim to guide local health officials and stakeholders, fostering better donor screening, stronger testing procedures, and lower rates of blood-borne disease transmission in the area.⁴

¹⁴**MATERIAL AND METHOD:**

Study Design:

This was a cross-sectional, descriptive study conducted from 1 March 2026 to 30 April 2026. A total 450 of sample of blood donor were screened during the specific period using a convenient sampling technique.¹ This objective of study was to find the prevalence of Hepatitis B and Hepatitis C among healthy donor during the specific period, proceeding directly after protocol clearance.¹³

Study Population:

The target population included all healthy voluntary, replacement and exchange donor aged 18-50 years, who visited the blood bank for blood donation during the specific period. And these all the donor fulfilled the selection criteria according to WHO and SBTG guidelines.¹

Study setting:

The study is conducted at Faisalabad,⁸ the third most populous city in Pakistan and the second largest city in the province of Punjab. Data was collected from the Jinnah and Mujahid hospital.

¹**Sample Collection and Processing:**

Donor's blood was drawn into the two blood collection evacuated tubes. First one was yellow top tube(SST) that contain gel for serum separation or second one was red top tube that contain no anticoagulant.

Inclusion Criteria:

It included all the participants aged above 16 and below 60, had a body weight exceeding 45kg, and hemoglobin concentration no less than 12.5g/dL. Vital sign of participants were

also noted such as systolic pressure and diastolic pressure between the ranges of 120/80mmHg, which confirming that the participants were healthy.

1 Exclusion Criteria:

Persons having a donation gap <3 months, drug abusers, marital status, sex workers, jaundice.

Ethical Consideration:

1 It was made sure that the donor information remained confidential and inaccessible to outsiders except for the research team.

Laboratory Testing:

Hepatitis B was detected by using HBsAg (hepatitis surface antigen) via ELISA technique. Hepatitis C was also detected by using via ELISA technique.

Result:

A total 450 donor samples were tested during this period for the hepatitis B and C. Among these donor sample, 287 (63.8%) were male, and 163 (36.2%) were female. And also these donor samples were divided into three groups 16-35 years, 35-55 years and greater than 55 years. Out of 450 sample, 107(23.8%) belonged to the group of 16-35 years, 212 (47.1%) belonged to the group of 35-55 years and the 131(29.1%) were above 55 years of age. (Table 1).

Demographic characteristics of Blood Donor:

Characteristics	Number(n)	Percentage
Total blood donor	450	100%
²³ Male	287	63.8%
Female	163	36.2%
Age 16-35 years	107	23.8%
Age 35-55years	212	47.1%
Age>55years	131	29.1%

Further analysis showed that between the 450 samples, Hepatitis B was found in 2.4% (n=11), the Hepatitis C was found in 13.6% (n=61) and the co-infection was found in 0.9% (n=4) which means that they were infected with both viruses. ⁵ Chi-square test applied. Statistically significant difference was observed between HBV, HCV prevalence and co-infection ($p < 0.001$). (Table 2).

² **Prevalence of Hepatitis B and Hepatitis C among Blood Donor:**

Type	Total blood donor	Positive cases	Percentage
Hepatitis B	450	11	2.4%
Hepatitis C	450	61	13.6%
Co-infection	450	4	0.9%
Total Infected	450	76	16.9%

Amid these 450 samples, HBV was more prevalent among male as compared to female. A total 10 male donor were found positive for HBV, giving the prevalence of 3.5%, on the other hand only one female was positive for HBV, giving the prevalence of 0.6%. And Hepatitis C virus infection, 39 male donors (13.6%) and 22 female donors (13.5%), were found positive for anti HCV antibodies. ²⁵ A statistically significant association was found between gender and HBV prevalence among blood donor ($p = 0.049$) with male donor showing a significantly higher prevalence (3.5%) as compared to female donor (0.6%). However, no significant was observed between gender and HCV prevalence ($p = 0.971$), as HCV was almost equally distributed among male (13.6%) and female (13.5%) donors. (Table 3)

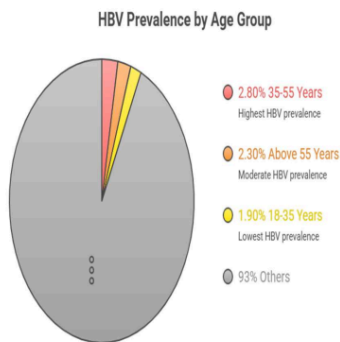
Gender-wise Distribution of HBV and HCV Infections:

Gender	Total number	HBV (+)	HBV prevalence	HCV(+)	HCV prevalence
Male	287	10	3.5%	39	13.6%
Female	163	1	0.6%	22	13.5%
Total	450	11	2.4%	61	13.6%

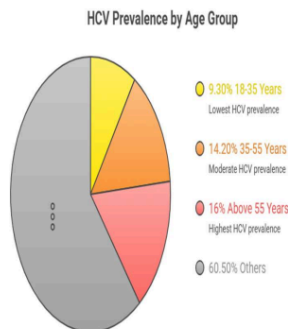
Among 450 blood donor samples, they were divided into three groups, and the highest prevalence of HBV was observed in group of 35-55 years, giving a prevalence of 2.8%, age group above 55 years, was observed giving the prevalence of 2.3%, while the lowest HBV prevalence was recorded in the youngest age group of 18-35 years, giving a prevalence of 1.9%. Hepatitis C virus (HCV) infection, a progressive increase in prevalence was observed with advancing age. The lowest HCV prevalence was noted in the 18-35 years age group, giving a prevalence of 9.3%, and the age group 35-55 years was also observed, giving the prevalence of 14.2%. The highest HCV prevalence was recorded in donors above 55 years of age, giving a prevalence of 16.0%. Age-wise analysis revealed no statistically significance association between age groups and HBV prevalence ($p=0.701$) or HCV prevalence ($p=0.143$). (Table 4)

Age-wise prevalence of HBV and HCV:

Age Group	Total	HBV(+)	HBV Prevalence	HCV	HCV Prevalence
16-35 years	107	2	1.9%	10	9.3%
35-55 years	212	6	2.8%	30	14.2%
>55 years	131	3	2.3%	21	16.0%
Total	450	11	2.4%	61	13.6%



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⁴Discussion:

The finding of the present cross-sectional study shed light on ²the burden of the HBV and HCV among blood donors in Faisalabad, Pakistan. It seems reasonable to conclude, that a total of 450 blood donors were screened, consisting of 287(63.8%) males and 163(36.2%) females. Out of which 11(2.4%) were found affirmative for HBV and 61(13.6%) were found ⁴positive for anti-HCV antibodies. Co-infection with both HBV and HCV was detected in 4(0.9%) ⁴donors. The prevalence of HBV, HCV and co-infection was statistically significant ($p<0.001$). There is some indication that the overall infection rate observed in this inquiry foregrounds the continued burden ²²of viral hepatitis among blood donors in this region.

The HBV prevalence of 2.4% recorded in the present study which is close to agreement with finding of (Zorob, Farooqi ⁶et al. 2023) who reported an HBV prevalence of 1.06% ²⁹among blood donor at a regional transfusion center in Punjab, Pakistan and noted that ¹Punjab province was reported as the most burdened for transfusion transmissible infections. Similarly (Sabir, Ghafoor et al. 2023) reported that HBV infections were most common in the age of 26-35 years in a Northern Pakistan transfusion center study.

The HCV prevalence of 13.6% observed in this study is significantly high and is consistent with the well-documented fact that Pakistan carries one of the biggest HCV burdens in the world. A large scale study conducted across Sindh, that prevalence of HBV and HCV showed a continuously rising trend between 2018 and 2022, with HBV positivity climbing from 1.46% in 2018 to 2.22% in 2022, and HCV rising from 1.25% to 1.66% over the same period and belonging to the 29-39 years age group. (Waheed, Gul et al. 2024). The local burden of unsafe injection practices, the reuse of surgical and dental instruments, a lack of community awareness and insufficient infection control measures that continue to fuel HCV transmission in central Punjab, and may be the reason for the relatively higher HCV prevalence in our Faisalabad-based study.

In terms of gender-wise distribution, the current study discovered that HBV prevalence was much greater in male donors (3.5%) as compared to female donors (0.6%) ¹⁰and this difference was found to be statistically significant ($p=0.049$, Fisher's Exact Test). The higher prevalence of HBV among male donors ²⁷in the present study is in accordance with earlier research and may be attributed to higher occupational stress, greater mobility, and higher risk behavior among men. HCV prevalence were nearly identical in both male (13.6%) and female (13.5%) and no statistically significant gender-based difference was observed ($p=0.971$). The nearly

equal distribution of HCV between genders indicates that HCV transmission in this region is not significantly influenced by gender. This might occur due to common community-level risk factors that affect both sexes equally such as tainted medical procedure, reuse of dental instruments and inadequate infection control practices rather than gender-specific behaviors.

With respect to age-wise distribution, the present study found that the 35-55 age group, had the highest HBV prevalence (2.8%), followed by the over 55 age group (2.3%), and the 18-35 age group (1.9%). However, this variation across age groups did not reach statistical significance ($p=0.701$). This pattern might be the result of older cohorts who were born before to Pakistan's broad adoption of HBV immunization programs having insufficient vaccine coverage. On the other hand, HCV prevalence in the current study clearly increased with age, peaking at 16.0% in donors over 55, ranging from 14.2% in the 35-55 age group and 9.3% in the 18-35 age group. Although this increasing trends did not achieve statistical significance ($p=0.143$), it remains clinically meaningful. The accumulative and prolonged exposure to HCV risk factors over time, such as parenteral exposures, unsafe blood transfusions, and dental procedures, which were particularly prevalent prior to the implementation of mandatory blood screening protocols in Pakistan, may account for the rising HCV trend with age seen in the current study.

In the current study, 4(0.9%) donors had co-infections with both HBV and HCV. Co-infection is clinically significant since it increases the risk of cirrhosis, hepatocellular carcinoma, and the rate at which liver disease progresses.

Recognizing this burden is more than just a transfusion issue; it is a mirror reflecting the community's larger viral hepatitis epidemic landscape. Every year, about 118.5 million people donate blood for pregnancy situations, emergencies, procedures, and charitable causes. Donating blood is an admirable way to save lives. For example, screening people for TTIs after roadside accidents with significant bleeding provides information about the prevalence and scope of specific diseases in the nation. Short-term infection during the first six months following infection is a characteristic of hepatitis B and hepatitis C infections. The severity might vary from a minor ailment with few or no symptoms to a serious condition that necessitates emergency hospitalization and critical treatment. One of the main ways that Hepatitis B and C are spread is through blood transfusions. (Zorob, Farooqi et al. 2023)

For Faisalabad, the larger Punjab region, the study's overall conclusions have significant public health ramifications. It is crucial to strictly enforce the requirement for blood donor

screening utilizing sensitive serological tests such as Chemiluminescent immunoassay and ELISA. Additionally, targeted awareness campaigns about the risks associated with unsafe injection practices and unsterilized medical equipment, the modes of transmission of HBV and HCV, and the availability of HBV vaccination should be regularly organized, especially among middle-aged and older population groups that demonstrated the highest HCV burden in this study. In order to lower the prevalence of viral hepatitis in Faisalabad, Pakistan, strengthening the EPI immunization program and raising infection control standards in community and healthcare settings continue to be crucial public health initiatives.

Conclusion:

The current study shows that among blood donors in Faisalabad, Pakistan, HBV and HCV infections continue to be a major public health concern. Co-infection was found in 0.9% of donors, with HBV and HCV prevalence of 2.4% and 13.6%, respectively. HBV was shown to be more common among male donors, but HCV increased steadily with age. These results emphasize the critical necessity for ongoing surveillance, required blood screening, and focused preventative measures. Community awareness initiatives, centralized blood collection systems, and the encouragement of unpaid volunteer donation are all highly advised. Strong governmental commitment, multisectoral cooperation, and a well-coordinated national surveillance system will ultimately be necessary to achieve a safe and hepatitis-free blood supply in Pakistan.

REFERENCE:

Almas, I., et al. (2024). "Molecular epidemiology and genetic characterization of hepatitis B virus in two major provinces of Pakistan." Archives of Virology **169**(11): 239.

Altaf, I., et al. "Prevalence and Risk Factors of Hepatitis B Virus Infection in a Private Tertiary Care Hospital in Kashmir: A Cross-sectional Study." Journal of Microscopy and Ultrastructure: 10.4103.

Arooj, K., et al. (2025). IDENTIFICATION AND ASSOCIATION OF RISK FACTORS FOR HEPATITIS B AND C IN FAISALABAD, PAKISTAN.

Arshad, S., et al. (2021). "Hepatitis B, C, and HIV Infections among blood Donors at Several Hospitals in Lahore, Pakistan." Journal of Pharmaceutical Research International **33**(55A): 112-126.

Aziz, S., et al. (2024). "Prevalence of HBV Infection and Associated Risk Factors in the General Population of District Peshawar, Pakistan." Indus Journal of Bioscience Research 2(02): 418-423.

Bhatti, M. M., et al. (2022). "The prevalence of transfusion transmitted infections among blood donors in Pakistan: A retrospective study." Oman Medical Journal 37(3): e386.

Ghazanfar, S., et al. (2022). "Frequency of transfusion transmissible infections among blood donors of Rawalpindi District, Pakistan." African Health Sciences 22(3): 590-598.

Iqbal, K., et al. (2026). "Prevalence and pattern of transfusion transmitted infections among blood donors visiting a tertiary care hospital, Pakistan." BMC Infectious Diseases.

Mustafa, Z. U., et al. (2021). "Effect of COVID-19 on viral hepatitis services in Pakistan." The Lancet Gastroenterology & Hepatology 6(3): 163-164.

Nazir, Q., et al. (2025). "Prevalence and risk factors of hepatitis B virus and hepatitis C virus among pregnant women of Faisalabad, Punjab, Pakistan." Journal of Medical & Health Sciences Review 2(2).

Niyomsri, S., et al. (2026). "The Impact of Hepatitis C and Socio-Demographic Variables on Health-Related Quality of Life in Pakistan: Cross-Sectional Study." Journal of Viral Hepatitis 33(4): e70139.

Sabir, N., et al. (2023). "Prevalence and Association of Transfusion-Transmissible Infections with Age of Blood Donors: A Regional Transfusion Centre Study in Northern Pakistan." J Coll Physicians Surg Pak 33(9): 978-982.

Saleem, U., et al. (2022). "Hepatitis C virus: Its prevalence, risk factors and genotype distribution in Pakistan." European Journal of Inflammation 20: 1721727X221144391.

Waheed, S., et al. (2024). "Prevalence and Trends of Hepatitis C, Hepatitis B, and Human Immunodeficiency Viruses Over Half a Decade Among Healthy Blood Donors Across Sindh, Pakistan." Cureus 16(4): e58374.

Waheed, U., et al. (2024). "The National Burden of Hepatitis C Among Blood Donors in Pakistan: A Systematic Review and Meta-Analysis (1996-2024)." Annals of Pims-Shaheed Zulfiqar Ali Bhutto Medical University 20(Suppl. 2): 848-863.

Waheed, U., et al. (2025). "The National Burden of Hepatitis B Among Blood Donors in Pakistan: A Systematic Review and Meta-Analysis (1991-2025)." Mirpur Journal of Medical Sciences 3(2): 71-86.

Zorob, T., et al. (2023). "Prevalence and trends in hepatitis B & C virus among blood donors in Pakistan: A regional transfusion center study." Livers 3(2): 271-281.

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